

Lab 04.1: Comfort indexes

03/10/2025

Abstract

This laboratory study investigates the vertical dynamics behavior of three distinct vehicles by utilizing a quarter car model. The primary focus is the evaluation of the discomfort index, defined as the standard deviation of the vertical acceleration of the sprung mass. The research independently analyzes how variations in five key mechanical parameters affect vehicle performance: unsprung mass, sprung mass, tire stiffness, suspension stiffness, and suspension damping.

Key Findings:

- **Mass Effects:** Increasing the unsprung mass leads to a slight increase in discomfort, though its overall influence is limited due to the distance between resonance peaks. Conversely, increasing the sprung mass results in a negative exponential decay of the discomfort index, which tends toward zero as mass increases.
- **Stiffness Effects:** Both tire stiffness and suspension stiffness show a positive correlation with discomfort. Higher values result in a linear increase in the index, while higher values contribute to decreased comfort by failing to isolate the vehicle from vibrations.
- **Damping Effects:** Unlike other parameters, the relationship between suspension damping and discomfort is non-monotonic. The analysis identifies a specific minimum point on the curve, representing the optimal damping level required to achieve maximum comfort.

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1 Introduction

The aim of this laboratory is to investigate the vertical dynamics behavior of different vehicles, using the quarter car model. The three vehicles taken into consideration have significantly different properties in terms of mass, tires, suspension stiffness and damping. The performance of the vehicle is evaluated by means of three indexes:

- Discomfort : it is associated with the vertical acceleration of the sprung mass
- Road Holding : it is associated with the vertical force exchanged between tire and road
- Working space : it is associated with the relative displacement between sprung and unsprung mass

They all represent standard deviations; therefore, larger values indicate poor performance. In this lab, only the discomfort has been taken into consideration:

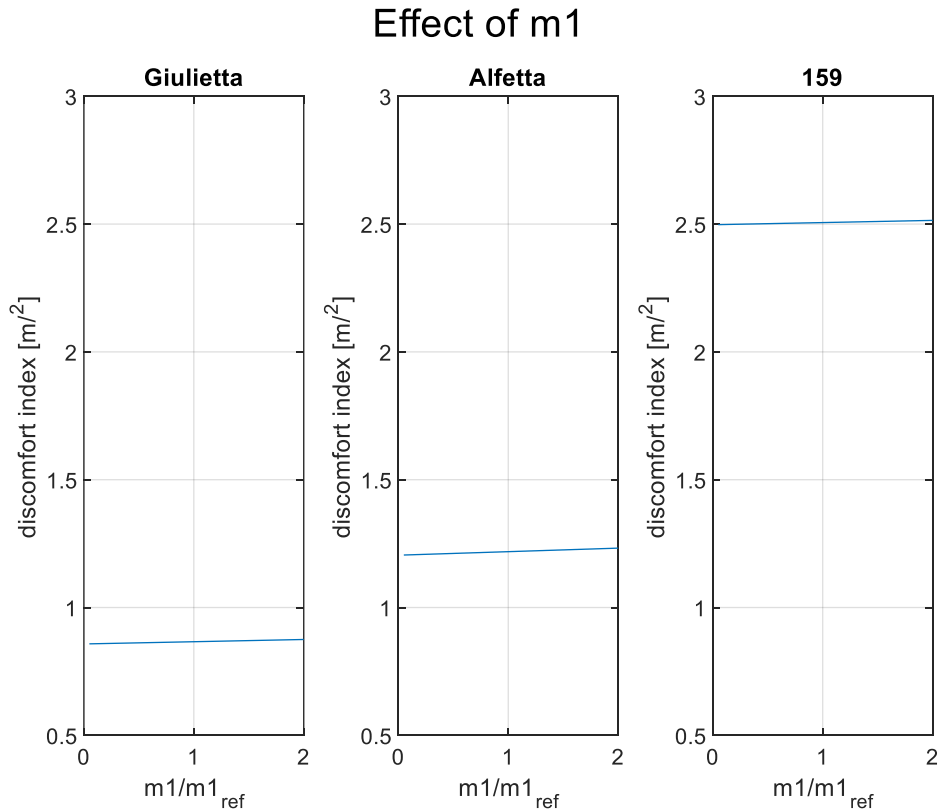
$$\begin{cases} \sigma_{\ddot{x}_2}^2 = \frac{1}{2} A_b v \bar{\sigma}_{\ddot{x}_2}^2 \\ \bar{\sigma}_{\ddot{x}_2}^2 = \frac{1}{m_2^2} \left(\frac{(m_2 + m_1) k_2^2}{r_2} + k_1 r_2 \right) \end{cases}$$

Figure 1

- $m_1 \rightarrow$ unsprung mass
- $m_2 \rightarrow$ sprung mass
- $k_1 \rightarrow$ equivalent tire stiffness
- $k_2 \rightarrow$ equivalent suspension stiffness
- $r_2 \rightarrow$ equivalent suspension damping
- $A_b \rightarrow$ road roughness index
- $v \rightarrow$ speed of the vehicle

The quarter car model parameters have been varied one at a time independently from each other.

2 Effect of m_1



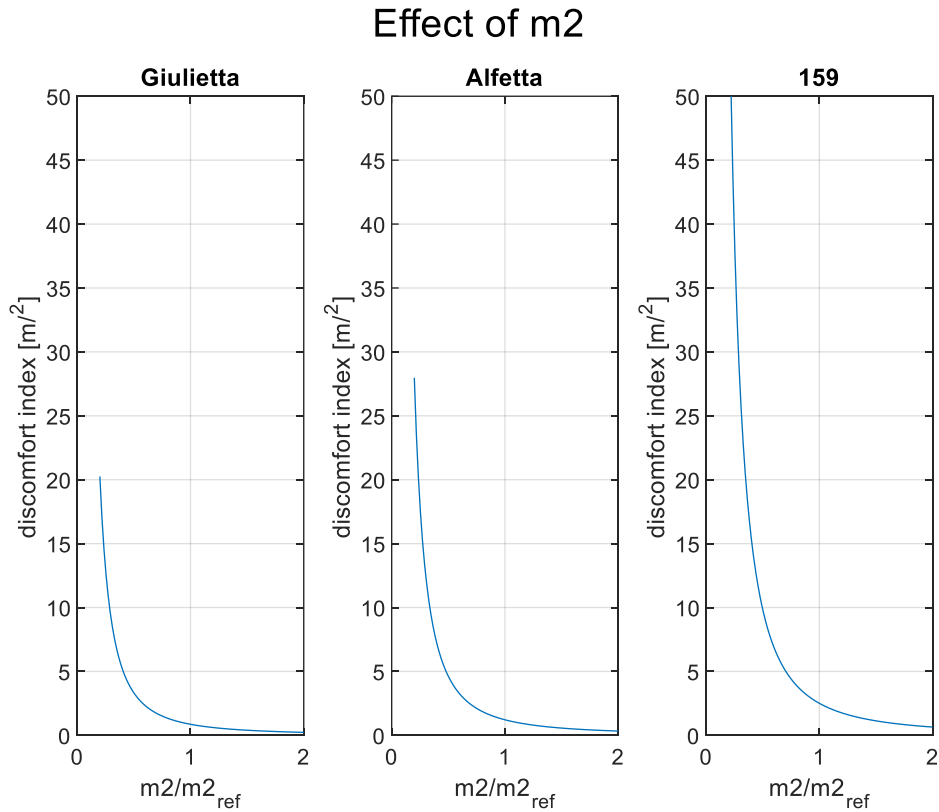
The presented plots illustrate the variation of discomfort index as a function of normalized unsprung mass.

It is observed that in all cases, the Discomfort Index exhibits a slight increase as increases.

This phenomenon is likely due to the fact that the Discomfort index measure the acceleration at the sprung mass.

Since the resonance peak of is sufficiently distant from that of , its influence on the overall discomfort is quite low.

3 Effect of m_2

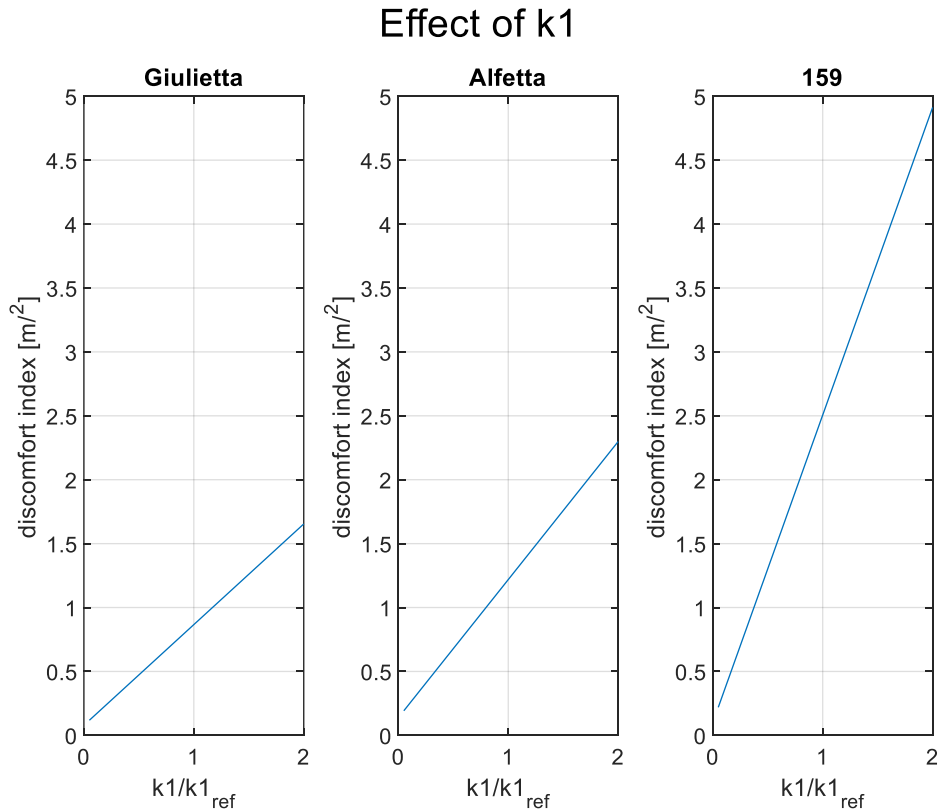


These plot represent the variation of discomfort index as a function of normalized sprung mass.

The curves shows a negative exponential decay as the sprung mass is incremented.

It can be observed that as m_2 approaches zero, the amplitude of the Discomfort index tends to infinite, otherwise, increasing the value of m_2 causes the discomfort index to drop down and tend to zero.

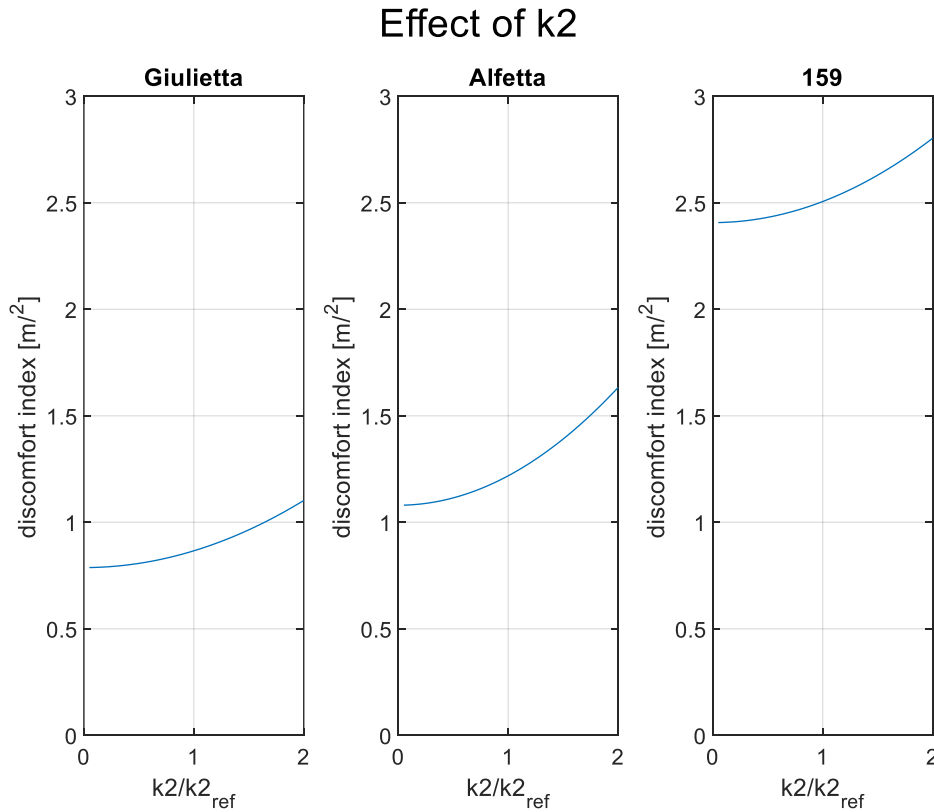
4 Effect of k_1



These plots represent the variation of discomfort index as a function of normalized stiffness of the tyre.

All cases demonstrate a linearly increasing relationship with respect to the increasing value of k_1 , and, by observing the formula used to compute the Discomfort index, it can be understood that a value of k_1 tending to infinity mathematically implies an infinite value of the Discomfort index.

5 Effect of k_2

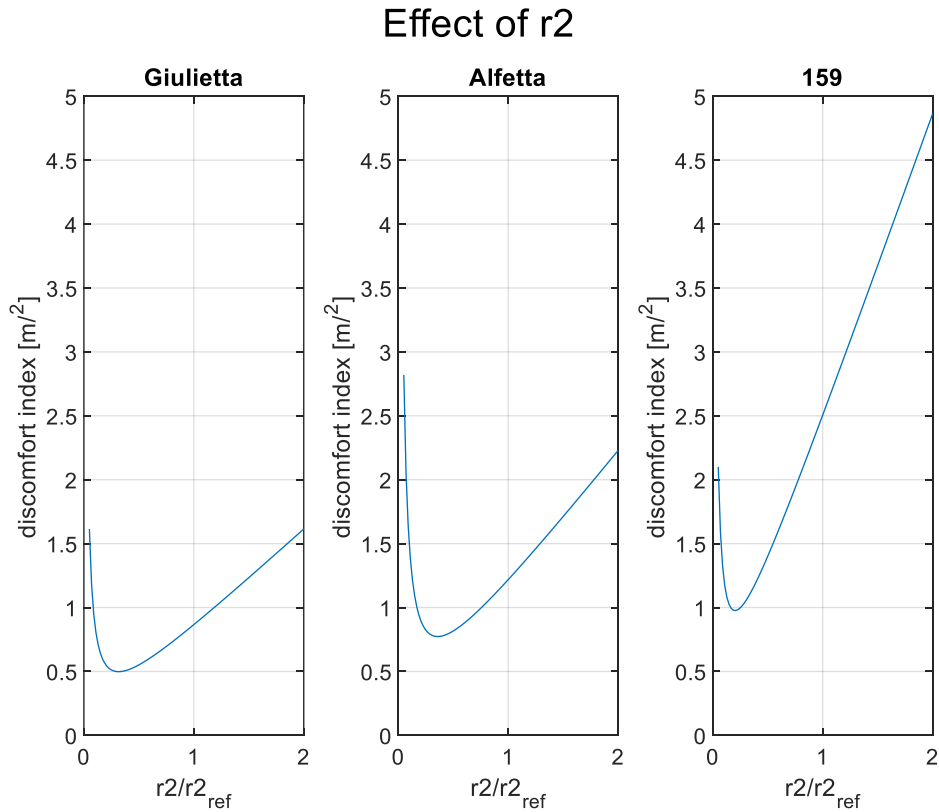


These plots represent the variation of the discomfort index as a function of the normalized stiffness associated with the suspensions. The general trend shows that as the normalized stiffness increases, so does the discomfort index.

By looking at the definition of the discomfort index itself (shown in Figure 1), it is straightforward that increasing k_2 , the standard deviation will increase and so the discomfort index.

Although the three curves show the same trend, the one associated to the 159 is evidently shifted upwards. This is due to the properties of the mechanical system itself, which is not able to isolate the car itself from the vibrations: the combination of a higher unsprung mass and suspension that are more stiff contribute together to this decrease of comfort.

6 Effect of r_2



The curves represented here describe how the discomfort index varies in relation to the normalized damping of the suspensions. Once again, the general trend remains constant for all the three models. Differently from the previous curves the shown trend is not monotonic: the curves show a minimum that implies a specific level of damping to have the best comfort.

7 Conclusion

This study demonstrates how mechanical parameters influence vehicle comfort within a quarter car model framework. While increasing the unsprung mass m_1 causes a slight rise in discomfort, higher sprung mass m_2 leads to a negative exponential decay in vertical acceleration. Instead, both tire and suspension stiffness show a positive correlation with discomfort, where higher values linearly degrade the vehicle's ability to isolate vibrations. Suspension damping r_2 follows a non-monotonic trend, identifying an optimal minimum point for maximum comfort. Concluding, these findings underline that while mass and stiffness generally follow linear or exponential trends, damping requires precise calibration to balance effectively the system's dynamic response.